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CIRCULATION

HAZARDOUS CHEMICALS AND THEIR HANDLING

Prepared

by

COMMITTEE

ON

HAZARDOUS CHEMICALS AND THEIR HANDLING

(Amended Edition - October 2002)

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NOTES

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Note 1 in superscript indicates the changes / modifications / additions as approved in 20th Safety Council Meeting held in October 2002.

FOREWORD

The Oil Industry in India is more than 100 years old. As such, a variety of practices have been in vogue because of collaboration / association with different foreign companies and governments. Standardisation in design philosophies and operating and maintenance practices at a national level was hardly in existence. This, coupled with feed back from some serious accidents that occurred in the recent past in India and abroad, emphasised the need for the industry to review the existing state-of-the-art in designing, operating and maintaining oil and gas installations.

With this in view, the Ministry of Petroleum and Natural Gas in 1986 constituted a Safety Council assisted by the Oil Industry Safety Directorate (OISD) staffed from within the industry for formulating and implementing a series of self regulatory measures aimed at removing obsolescence, standardising and upgrading the existing standards to ensure safer operations. Accordingly, OISD constituted a number of technical committees comprising of experts nominated from the industry to draw up standards and guidelines on various subjects.

The present document “Standard on Hazardous Chemicals and their Handling” was prepared by the Committee on “Hazardous Chemicals”. This document was prepared based on the accumulated experience and knowledge of industry members, and various national and international codes and practices.

This document will be reviewed periodically for improvements based on the new experiences and better understanding. Suggestions from industry members may be addressed to:

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HAZARDOUS CHEMICALS AND THEIR HANDLING

1.0 INTRODUCTION

In the Petroleum and Natural Gas Production and Processing Industry, a number of hazardous chemicals are required to be stored and handled in appreciable quantities. In view of the stringent product quality requirement, various complex processes are introduced involving the handling of hazardous chemicals. Some additives are sometimes introduced to meet the quality requirement of various petroleum products. These chemicals pose various types of hazards like flammability, toxicity, explosivity, corrosivity etc.

Inadequate awareness about the hazardous properties of these chemicals may lead to serious accidents which will affect the men at work and the environment . The present standard will help in facilitating the safe storage and handling of hazardous chemicals and in reducing the number of accidents and occupational diseases.

2.0 SCOPE

This standard deals with the hazardous chemicals used in Petroleum and Natural Gas production, processing and transportation facilities including those listed in The Manufacture, Storage and Import of Hazardous Chemicals Rules 1989 and The Factories Act. This standard does not include hazardous chemicals namely chemical dust, laboratory chemicals, proprietary chemicals and radioactive materials. The safety data sheets appended as schedule 1 have been prepared as per the proforma given in Schedule 9 of Manufacture, Storage and Import of Hazardous Chemicals Rules, 1989.

3.0 DEFINITION

3.1 SYNONYMS: For common chemicals, several chemical names and numerous trade names may be applied to describe the chemical in question In synonyms field, many of these names are identified to aid users on the range of names which have been used to describe each substance.

3.2 CAS NO. : Chemical Abstracts Service Registry Number is a numeric designation assigned by American Chemical Society's Chemical Abstracts Service and uniquely identifies a specific chemical compound. This entry allows one to conclusively identify a material regardless of the name or naming system used.

3.3 UN NUMBER: The united nations number is a four figure code used to identify hazardous chemicals and is used for identification of chemicals transported internally by road, rail and by air.

3.4 Hazardous Waste ID. No. : This means the categories of wastes specified in the schedule in The Hazardous Waste (Management and Handling) Rules, 1989

3.5 FLAMMABLE CHEMICALS

Flammable Gases : Chemicals which in the gaseous state at normal pressure and mixed with air become flammable and the boiling point of which at normal pressure is 20°C or below.

Highly Flammable Liquids : Chemicals which have a flash point lower than 23 °C and boiling point of which at normal pressure is above 20 °C.

3.6 FLAMMABLE (EXPLOSIVE) LIMIT AND RANGE

In the case of gases or vapour, which form flammable mixtures with air, there is a minimum concentration of vapour in air below which the propagation of flame does not occur in contact with a source of ignition. This is called Lower Explosive Limit (LEL). Similarly, there is a maximum concentration of vapour above which the propagation of flame does not occur on contact with a source of ignition. This is called Upper Explosive Limit (UEL). This range of minimum to maximum concentration is termed as Flammable (Explosive) range.

The flammable limits are not appreciably changed by normal variations in atmospheric pressure and temperature. However, at high temperatures, the upper limit is raised and the lower limit lowered, resulting in greater flammable or explosive range. The effect of high pressures on the limits is different for each gas or vapour depending upon its nature.

3.7 TDG FLAMMABILITY

Transport of dangerous goods flammability is used in Canada and is based on Classification of dangerous goods by UN Committee of experts on transport of danger goods. The classes indicated are 2, 3, 4 and 5 which are same as indicated in The Central Motor Vehicle Rules 1989.

3.8 FLASH POINT

The minimum temperature at which the liquid gives so much vapour that this vapour, when mixed with air, forms an ignitable mixture and gives a momentary flash on application of a small pilot flame under specified conditions of test.

3.9 AUTO IGNITION TEMPERATURE

The lowest temperature at which ignition occurs in a mixture of explosive gas and air when the method specified in IS: 7820-1975 is followed.

3.10 EXPLOSIVES

Chemicals which may explode under the effect of flame, heat or photo chemical condition or which are more sensitive to shocks or friction than Di-Nitro Benzene.

3.11 CORROSIVE CHEMICALS

Chemicals, which will cause damage to the living tissue and/or chemically attack metallic containers and structures are corrosive chemicals.

3.12 OXIDISING SUBSTANCES (OXIDISER):

Oxidising substances means a chemical that initiates or promotes combustion in other materials, thereby causing fire either of itself or through release of oxygen or other gases.

3.13 PYROPHORIC SUBSTANCES

Chemicals which are so reactive that on contact with air and its moisture, oxidation/hydrolysis occurs at such a rate so as to cause ignition. Some of these reactions liberate flammable gases.

3.14 ORGANIC PEROXIDES

Organic peroxide means an organic compound that contains the bivalent –o-o– structure and which may be considered to be a structural derivative of hydrogen peroxide where one or both of hydrogen atoms has been replaced by an organic radical.

3.15 TOXIC CHEMICALS

Chemicals having the following values of acute toxicity and which owing to their physical & chemical properties, are capable of producing major accident hazards :

Medium lethal dose by oral route (oral toxicity) LD50 (mg/kg body weight of test animals)	Medium lethal dose by the dermal route (dermal toxicity)LD50 (mg/kg body weight of test animals)	Medium lethal concentration by inhalation route (four hrs.) LC50 (mg/l inhalation in test animals)
1. Extremely toxic 1-50	1-200	0.1-0.5
2. Highly toxic 51-500	201-2000	0.5-2.0

3.16 INCOMPATIBLE CHEMICALS

These are chemical substances which in contact with each other cause hazardous reactions which may be violent or produce toxic materials.

3.17 TLV (THRESHOLD LIMIT VALUE)

The term refers to concentrations of air contaminants in the working environment to which it is believed, nearly all workers may be exposed repeatedly day after day without adverse effect. Normally, it is the time weighted average concentration in ppm or gm/m^3 for a normal 8-hour work day or 40 hour workweek. Excursions above this limit may occur if they are compensated during the workday, by equivalent excursion below the limit.

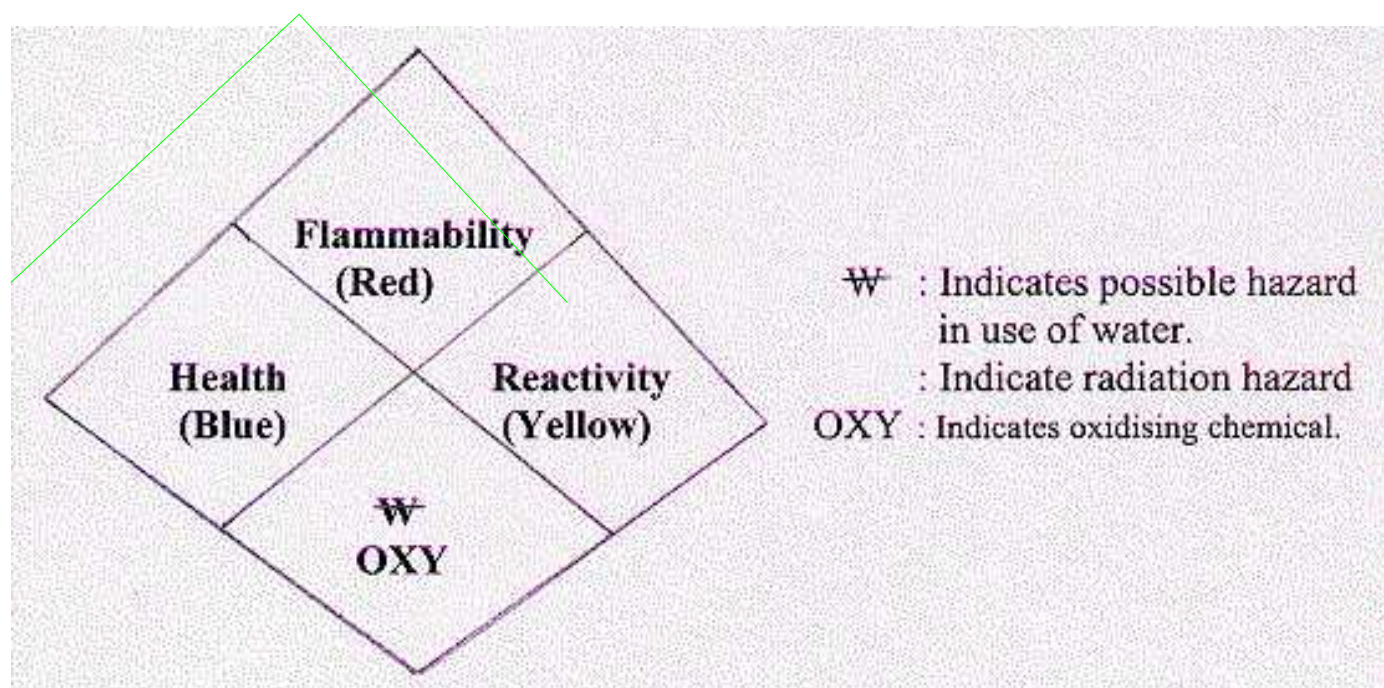
3.18 STEL (SHORT TERM EXPOSURE LIMIT)

It is defined as a 15 minutes time weighted average exposure, which should not be exceeded at any time during a workday even if the 8 hours time weighted average is within the TLV. Exposure at the STEL should not be longer than 15 minutes and should not be repeated more than four times per day. There should be at least 60 minutes interval between successive exposures at the STEL. An averaging period other than 15 minutes may be recommended when this is warranted by observed biological effects.

4.0 EXPLANATORY NOTES

4.1 NUMBERING THE NFPA PICTORIAL REPRESENTATION OF HAZARD

NFPA pictorial representations consist of a diamond as shown in the figure below, divided into 4 parts. These parts, starting from left block (in clockwise direction), represent Health Hazard, Flammability Hazard, Reactivity Hazard and Space for additional information such as water reactivity, oxidant and radiation hazard. The colour codes for these hazards are : blue for health, red for flammability, yellow for reactivity and colourless for additional information. The intensity of hazard is grouped into five classes with numerical indications as 0, 1, 2, 3 & 4 in the ascending order of hazard intensity. In the following lines, these hazard intensities are explained, with the numeral on the left indicating the hazard intensity.



4.1.1 HEALTH

4 A few whiffs of the gas or vapour could cause death or the gas, vapour or liquid could be fatal on penetrating the fire fighters normal full protective clothing which is designed for resistance to heat. For most chemicals having a Health 4 rating, the normal full protective clothing available to the average fire department will not provide adequate protection against skin contact with these materials. Only special protective clothing designed to protect against the specific hazard should be worn.

3 Materials extremely hazardous to health, but areas may be entered with extreme care. Full protective clothing, including self- contained breathing apparatus, rubber gloves, boots and bands around legs, arms and waist should be provided. No skin surface should be exposed.

2 Materials hazardous to health, but areas may be entered freely with self-contained breathing apparatus.

1 Materials only slightly hazardous to health. It may be desirable to wear self-contained breathing apparatus.

0 Materials which on exposure under fire conditions would offer no health hazard beyond that of ordinary combustible material.

4.1.2 FLAMMABILITY

4 Very flammable gases, very volatile flammable liquids, and materials that in the form of dusts or mists readily form explosive mixtures when dispersed in air. Shut off flow of gas or liquid and keep cooling water streams on exposed tanks or containers. Use water spray carefully in the vicinity of dusts so as not to create dust clouds.

3 Liquids, which can be ignited under almost, all normal temperature conditions. Water may be ineffective on these liquids because of their low flash points.

Solids which form coarse dusts, solids in shredded or fibrous form that create flash fires, solids that burn rapidly, usually because they contain their own oxygen, and any material that ignites spontaneously at normal temperatures in air.

2 Liquids which must be moderately heated before ignition will occur and solids that readily give off flammable vapours. Water spray may be used to extinguish the fire because the material can be cooled to below its flash point.

1 Materials that must be pre-heated before ignition can occur. Water may cause frothing of liquids with this flammability rating number if it gets below the surface of the liquid and turns to steam.

However, water spray gently applied to the surface will cause a frothing which will extinguish the fire. Most combustible solids have flammability rating of 1.

0 Materials that will not burn.

4.1.3 REACTIVITY

4 Materials which in themselves are readily capable of detonation or of explosive decomposition or explosive reaction at normal temperatures and pressures. Includes materials, which are sensitive to mechanical or localized thermal shock. If a chemical with this hazard rating is in an advanced or massive fire, the area should be evacuated.

3 Materials which in themselves are capable of detonation or of explosive decomposition or of explosive reaction but which require a strong initiating source or which must be heated under confinement before initiation. Includes materials which are sensitive to thermal or mechanical shock at elevated temperature and pressure or which react explosively with water without requiring of confinement. Fire fighting should be done from an explosion-resistant location.

2 Materials which in themselves are normally unstable and readily undergo violent chemical change but do not detonate. Includes materials which can undergo chemical change with rapid release of energy at normal temperatures and pressures or which can undergo violent chemical change at elevated temperatures and pressures. Also includes those materials which may react violently with water or which may form potentially explosive mixtures with water. In advanced or massive fires, fire fighting should be done from a protected location.

1 Materials which in themselves are normally stable but which may become unstable at elevated temperature and pressure or which may react with water with some release of energy but not violently. Caution must be used in approaching the fire and applying water.

0 Materials which are normally stable even under fire exposure conditions and which are not reactive with water. Normal fire fighting procedures may be used.

Explanation for bottom space (Colourless) : The bottom space is primarily used to identify unusual reactivity with water. With a line through its centre alerts fire fighting personnel to the possible hazard in use of water. This bottom space may also be used to identify a radiation hazard by the symbol. Oxidising chemicals are identified in the bottom space, as OXY.

4.2 HAZCHEM CODE: This is an emergency action code for fire and spillage. The significance of the letters and the numbers indicated in this code are as under:

**EMERGENCY ACTION
CODE SCALE
FOR FIRE OR SPILLAGE**

- 1: JETS
- 2: FOG
- 3: FOAM
- 4: DRY AGENT

P	V	FULL	DILUTE
R			
S	V	BA	
S		BA for FIRE only	
T		BA	CONTAIN
T		BA for FIRE only	
W	V		
X		FULL	
Y	V	BA	
Y		BA for FIRE only	
Z		BA	
Z		BA for FIRE only	
E		CONSIDER EVACUATION	

Notes for Guidance :

FOG : In the absence of fog equipment a fine spray may be used.

DRY AGENT : Water must not be allowed to come into contact with the substances at risk.

V : Can be violently or even explosively reactive.

FULL : Full body protective clothing with BA.

BA : Breathing apparatus plus protective gloves.

DILUTE : May be washed to drain with large quantities of water.

CONTAIN : Prevent, by any means available, the spillage from entering drains or water course.

SUBSTANCES	HAZCHEM CODE
1) PETROL	3 Y E
2) METHYL ALCOHOL	2 PE
3) L.P.G.	2 WE
4) AMMONIA LIQUEFIED	2 PE
5) SULPHUR	2 Z
6) KEROSENE	3 Y

5.0 CLASSIFICATION OF HAZARDOUS SUBSTANCES

The hazardous substances are classified into eight classes. These are as under :

CLASS 1 EXPLOSIVES

CLASS 2 GASES, COMPRESSED LIQUEFIED
DISSOLVED OR DEEPLY REFRIGERATED, INFLAMMABLE GASES, POISON
(TOXIC) GASES.

CLASS 3 INFLAMMABLE LIQUIDS

CLASS 4 INFLAMMABLE SOLIDS, SUBSTANCES LIABLE TO
SPONTANEOUS COMBUSTION; SUBSTANCES WHICH ON CONTACT WITH
WATER EMIT INFLAMMABLE GASES.

Division 4.1 Inflammable solids

Division 4.2 Substances liable to spontaneous
combustion.

Division 4.3 Substances, which in contact with water, emit
inflammable gases.

CLASS 5 OXIDISING SUBSTANCES

Division 5.1 oxidizing substances

Division 5.2 Organic peroxides

CLASS 6 POISONOUS (TOXIC) AND INFECTIOUS
SUBSTANCES

Division 6.1 Poisonous (toxic) substances

Division 6.2 Infectious substances

CLASS 7 RADIOACTIVE SUBSTANCES

CLASS 8 CORROSIVES

For details of their pictorial representation, the Central Motor Vehicles Rules 1989 may be referred.

6.0 GENERAL GUIDELINES FOR SAFE STORAGE, HANDLING AND DISPOSAL ^{Note 1}

6.1 AWARENESS OF OPERATING STAFF

Safety at storage and handling of hazardous chemicals depends on a considerable extent on effective safety education / training following safety procedures and efficient supervision. The operating staff must be aware of the hazards involved, the location and operation of safety equipment.

6.2 STORAGE OF HAZARDOUS CHEMICALS

6.2.1 STORAGE IN BULK

Utmost care is necessary in ensuring proper and safe storage facility for hazardous chemicals. The following points should be given due weightage to avoid any mishap.

- (i) Attempt should be made to find suitable less hazardous alternate chemicals, to replace the hazardous chemical. The inventory of all hazardous chemicals for that matter must be kept as minimum as possible.
- (ii) The tanks should be located so as not to pose safety problems due to leakage and reaction with other chemicals stored nearby. The environment at the location must be compatible with the chemical stored.

The storage area should be declared as a prohibited area and should be provided with fencing having at least two exits / “No Smoking” and/or “Prohibited Area” display boards, as applicable should be provided at site.

- (a) Oxidising Agents : Oxidising substances must be stored away from all flammable materials even if they are only slightly flammable. Oxidising material must be kept away from substance which are reducing agents.
- (b) Water Sensitive Substances : These substances react with water or steam to produce flammable or explosive gases and evolve heat, eg.. Conc. acids like sulphuric acid, glacial ethnoic acid etc. Conc. Alkali like sodium hydroxide react with water later to evolve heat. Such materials must not be stored in areas where water flooding from pipe leakages or leaky roofs can happen.

- (c) Toxic Substances : These substances must be stored in well ventilated areas, preferably cool and certainly away from direct sun rays. Periodic checks should be organised for all parts of storage areas. Extremely toxic substances like TEL (Tetra Ethyl Lead) must be stored separately under strict security.
 - (d) Incompatible Chemicals : Incompatible chemicals should not be stored near each other.
 - (e) Corrosive Chemicals : The flooring of the area where corrosive chemicals are stored, shall be impervious and made of corrosion resistant materials.
- (iii) The storage tank and foundation should be of suitable material of construction to prevent corrosion.
 - (iv) The connections and openings to the tank should be as minimum as possible so that the possibility of leakage and maintenance hazards are minimized.
 - (v) The tanks should be fitted with vents/relief systems of adequate capacity discharging to a safe height. Capacity requirement in case of high vapour evolution resulting from heating due to fire of adjacent tank also should be considered. Wherever atmospheric release of the vapour is not advisable, the relief should be routed to a neutralising system. Flame arrestors are recommended for the vents of atmospheric storage tanks containing Class A (flash point less than 23 degree centigrade) chemicals. The location of flame arrestor must be easily accessible for periodic inspection to ensure that flame arrestor is free of any choking.
 - (vi) The tank should have provision for emergency evacuation of its stock to a safer location in case of emergency like fire, tank failure etc. using suitable pump out system. Use of double isolation valve – gate and globe valve in series on the drain outline is recommended.
 - (vii) Each storage tank should have necessary instruments to monitor its level, pressure and temperature preferably with remote indications at control room or field operators cabins. Abnormal rise in level / pressure / temperature should alert control room by alarm. Quite often, the level gauges pose problems of choking, leakage from drain connections, leakage from broken gauge glasses leading to accidents. Hence, it is suggested to have guarded, illuminated level gauges. Level gauges should have provision for in-situ cleaning.

Wherever possible any leakage of hazardous chemicals must be detected and alarm should be annunciated at control room.

(vii) The tank should have a dyke of suitable material (compatible with the chemical) of volume equal to the volume of the larger storage tank. The dyke should have facility to:

- (a) drain off rain water into storm water channel ;
- (b) route high volume spillage / leakage to suitable neutralizing pit nearby ;
- (c) discharge safe effluent to oily water system, as applicable.

The isolation valves on dyke drains should be located outside the dyke.

- (ix) Fire access roads should be provided to storage area. The storage tanks / area should have suitable fire protection and fire fighting facility.
- (x) Adequate communication facility like public address / telephone must be made available for interacting with control room, fire stations, medical units etc.
- (xi) The name of chemical, type of hazard, emergency operational instructions, antidote first aid etc. should be displayed near each tank.
- (xii) Electrical Safety : All cables and electric fittings shall be constructed, installed, protected, operated and maintained in such a manner so as to prevent risk of open sparking.

6.2.2 STORAGE OF CHEMICALS IN DRUMS / OTHER CONTAINER

This section has to be read in continuation with previous section (5.2.1) as most of the above guidelines for bulk storage apply to drums / others containers storage. In fact, drum storage of hazardous chemicals (liquids) pose potential hazard compared to bulk storage in that (a) each drum of a cluster of drum can become a source of leakage; (b) the drums normally being less resistant to fire would collapse faster escalating and spreading of fire to other drums of the stack. Hence following guidelines Shall be considered in addition to those discussed in 5.2.1 :

- (i) The drums should never be filled full with the liquid chemical. There should be sufficient ullage to take care of thermal expansion.
- (ii) The drums should preferably be stored in a well ventilated shed (preferably away from process units) with impermeable floor sloping away from drums.
- (iii) There should be provision to collect accidental spills for safe disposal.
- (iv) The drums should be stacked in a single tier with the bung (lid) up in an orderly manner so that all parts of the storage space are accessible.

- (v) Empty drums should be removed away and made free of its contents before being kept in safe place.
- (vi) Periodic site inspection should be carried out to ensure that there is no leakage from any of the drums.
- (vii) Where combustible solid materials are stored, the dust content of the air must be kept below the lower explosive limit. Efficient dust collection system and good house keeping should be ensured.

6.2.3 STORAGE OF COMPRESSED GAS CYLINDERS

The nature of chemical gas apart, the high pressure of storage amplifies the hazard of the storage gas cylinders. Following are the recommended guidelines for storage of cylinders :

- (i) Cylinders should be protected adequately from variation in temperatures. They should be stored in cool, dry, well ventilated and covered place away from steam pipes, furnace, boilers or any other potential source of heat. Cylinder should not be kept near gangways / elevators to avoid any hard object striking them.
- (ii) Cylinders containing flammable gases and toxic gases shall be kept separated from each other and from cylinders containing other types of gases by an adequate distance or by a suitable partition wall.
- (iii) Cylinders should be properly stacked taking care to prevent knocking, dropping or rolling of cylinders.
- (iv) Cylinder nozzle should be suitably protected against damage. Oil or similar lubricant should not be used on valves or other fittings of cylinders containing gas.
- (v) In case of liquefied gas like liquid nitrogen, the cylinders should be stacked vertical so that in case of defective regulator only gas would escape.
- (vi) The cylinder valves should be provided with security cap on the outlet to act as a secondary means of safeguard. If there is a leak in the valve which cannot be rectified by tightening the gland nut or spindle, the cylinder should be removed to an open space where it is less dangerous to life & property and necessary remedial measures should be taken.
- (vii) It is necessary that a) name of chemical ; b) state of chemical as to gas or liquid ; c) pressure of the contents ; d) date of filling and e) name and address of the source (manufacturer) should be labelled on the cylinders.

- (viii) The due date of test for cylinder is should be marked on the metal ring inserted between the cylinder valve and cylinder neck. If this date is over, the cylinder should not be accepted.
- (ix) For inflammable gases, the cylinder storage area should be made of non combustible materials and flame proof fittings should be used.
- (x) The floor level at storage area should be sufficiently above ground level to prevent water logging and corrosion. Flooring of the storage area should not be wet or muddy.
- (xi) Filled and empty cylinders should be tagged and kept separately.
- (xii) The storage area should be fenced and marked as Prohibited Area with NO SMOKING WARNINGS and should have at least two exits.
- (xiii) Periodic inspection of cylinders for ascertaining leakage should be made. Automatic gas detectors where applicable should be provided at suitable points so that in case of leakage, alarm is hooted at control room.
- (xiv) Emergency kit, safety protective equipment and clothing should be available in close vicinity to storage area.
- (xv) Adequate and suitable fire extinguishers must be available at site.
- (xvi) For proper identification of the chemical all cylinders shall have the proper colour as Rule 8 of The Gas Cylinders Rules 1981.

6.3 HANDLING OF HAZARDOUS CHEMICALS

6.3.1 UNLOADING OF TANK TRUCKS / TANK WAGONS

- (i) Before the tanker enters the industry premises, the tanker is to be inspected for authorized entry and safe & sound condition of the tanker, its contents and that of the prime mover. Tankers entering plant are to be fitted with flare arresters on their exhaust.
- (ii) The quality of the chemical in the tanker should be ascertained before unloading to avoid contamination of chemical already at storage.
- (iii) In case of flammable chemicals, the prime mover (engine) should be kept of. The tanker should be properly blocked from movement before connections are made for unloading hazardous chemicals.

- (iv) In case of flammable chemicals, the unloading point should be located at a safe distance outside the storage dyke. For details, refer OISD-STD-118 on Layout.
- (v) Pressurizing with air / inert gas for unloading should be avoided. It is recommended to use pumps / vacuum systems for unloading. Pumps should preferably be of seal less type and valves should be of glandless types. Solid chemicals in bulk should be handled with lifting machines and conveyors.
- (vi) Coupling used for connecting hose to tanker must be leak proof. Flange connections are preferred. Where threaded connections are used, the threaded portion should be properly preserved against corrosion / wearing of threads and thoroughly inspected before connections are made.
- (vii) The unloading hose should be devoid of cracks & blisters and should be capable of withstanding whatever pressure developed during unloading operation. The hose should be hydro-tested at a frequency guided by experience. Proper records of hydro-test should be maintained.
- (viii) Same hose should not be used for unloading different chemicals. Hoses for different chemicals should be marked with different colour stripes for easy identification.
- (ix) For flammable chemicals, the tanker and the hose are to be properly earthed before starting unloading operation.
- (x) Unloading should preferably be done in day time.
- (xi) Unloading should be done under personal supervision of responsible staff authorized by the management.
- (xii) The operating staff must use suitable personal protective clothing / equipment. Suitable breathing canisters and first aid box must be available at site for use in case of emergency.
- (xiii) Provision of sample quantity of water / neutralizing medium to take care of leakage / spillage must be made. Also steam and inert gas hose stations must be available at unloading point.
- (xiv) There must be adequate illumination at site. Flame proof fittings should be used wherever necessary.
- (xv) The unloading systems should have facility to vent / drain the remaining chemical in the hose to a suitable safe point. The hose should be kept blinded when hot in use. Thermal safety valve discharging to safe disposal or handling facility should be provided.

- (xvi) Before starting unloading, the ullage of the receiving tank should be checked. Care should be taken to avoid overflow of tanks. Gas / chemical leak detection system to sound an alarm at the control room/site may be provided wherever possible, so that quick remedial measures can be taken. Wherever necessary, quick/remote isolation valves should be provided.
- (xvii) Fire alarm and fire fighting facility commensurate with the chemical – as mentioned in the data sheet (Chapter 6) should be provided at the unloading point.
- (xviii) Effective communication system like public address / telephone must be available for communicating with the control room / fire station / health unit.

6.3.2 UNLOADING OF DRUMS / CONTAINERS

This section should be read in addition to Section 5.3.1 for general precautions

- (i) Manual handling of drums / containers should be minimized. It is preferable fork-lifters and suitable cradles are used to handle drums.
- (ii) Carboys containing hazardous chemicals should not be subjected to impact.
- (iii) Suitable protective clothing should be used while handling drums / containers and the operators should position himself such that he is in the upwind direction so that even in case of accidental release of chemical, he is safe.

6.3.3 HANDLING OF CYLINDERS

This Section is to be read in addition to Section 5.3.1 and 5.3.2 for certain common guidelines :

- (i) The cylinder should be checked for proper markings as listed below, if not in order the cylinder should be returned to the supplier.
 - (a) Owner's / manufacturer's name.
 - (b) Specification to which cylinder has been made.
 - (c) Date of last hydrostatic test.
 - (d) Working pressure and test pressure.
 - (e) Tare weight.

- (f) Water capacity and proper identification colour of the chemical.
- (ii) Cylinders should be unloaded using fork-lifter, hoist etc. Manual handling is not recommended. Valve hoods should not be used as supports while unloading.
- (iii) Ensure the cylinder is not leaking from valve / body. Ensure that the cylinder is not dropped / bumped against any other cylinder or hard object.
- (iv) For increased flow rate of gas, suitable vaporizer can be used. Direct heating of the cylinder with steam or flame shall not be done.
- (v) Pressure gauge and flow measuring device should be used to monitor flow.
- (vi) Use non return valve to prevent back flow of process fluid into the partially empty / empty cylinder.
- (vii) When the cylinder is not required to be used for a considerable period of time, it should be disconnected from process and kept completely closed, at storage, marked accordingly.
- (viii) Once a cylinder is consumed, close the cylinder valve fully first and then the process valve. Gently remove the connection, put an 'Empty' label on the cylinder and store at the location specified for empty cylinders.

6.3.4 SAFE DISPOSAL OF EMPTY CONTAINERS OF HAZARDOUS CHEMICALS Note 1

Safe disposal of empty containers of hazardous chemicals should be done based on guidelines given by the Vendors.

7.0 HAZARDOUS CHEMICALS HANDLED IN OIL AND GAS INDUSTRY

This section includes safety data sheets of most of the hazardous chemicals being used in Petroleum Industry. The data provided are compiled from various data sources as given in Reference. The identity of the chemical is defined in para 1 of the safety data sheet and important physical and chemical properties of each chemical have been given in para 2 of the safety data sheet. The fire & explosion data, reactivity data, health hazardous data have been given vide para 3, 4 & 5 respectively. The preventive measures and emergency & first aid

measures and additional informations/references, if any, are given in para 6,7& 8 respectively. Individual data sheets are to be supplemented with the following as item 9 and 10.

9. MANUFACTURER'S / SUPPLIER'S DATA

Name of Firm	Contact Person in Emergency
Mailing Address Telephone/Telex Nos.: Telegraphic Address :	Local Bodies involved.
	Standard Packing Trem Card Details / Ref.
	Other

10. DISCLAIMER

Information contained in this material data sheet is believed to be reliable but no representation, guarantee or warranties of any kind are made as to its accuracy, suitability for a particular application or results to be obtained from them. It is upto the manufacturer/seller to ensure that the information contained in the material safety data sheet is relevant to the product manufactured or sold by him as the case may be. OISD makes no warranties expressed or implied in respect of the adequacy of this document for any particular purpose.